Course Project - AE706

Vorticity-Velocity Stream function formulation solver for skewed cavity

23B0046

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Introduction:

In this project, a vorticity-stream function based solver is programmed for a skewed domain. The grid transformations are listed below along with the differential equation and boundary conditions in both original and transformed domains. Finally results are compared against the reference paper and are shown to be closely matching.

Grid Transformation

An affine transformation to the physical domain is applied to convert it to a rectangular area in the computational domain.
The mapping from the computational domain (ξ, η) to the physical domain is as follows:

$$x(\xi, \eta) = \xi + \eta \cos(\theta)$$
$$y(\xi, \eta) = \eta \sin(\theta)$$

Here, θ is the skew angle of the cavity. Is is assumed that the base of the cavity aligns with the $y = 0$ line.
This isn't a very large constraint as any domain arrived by an affine transformation from a rectangle can be rotated to have it's bottom edge align with the $y = 0$ line.
And as such, we can also construct the inverse transformations

$$\xi(x, y) = x - y\cot(\theta)$$
$$\eta(x, y) = \frac{y}{\sin(\theta)}$$

From this, we can also find the derivatives of each component with respect to the other components.

$$
\begin{aligned}
\xi_x &= 1 \\
\xi_y &= -\cot(\theta) \\
\eta_x &= 0 \\
\eta_y &= \frac{1}{\sin(\theta)} \\
x_\xi &= 1 \\
x_\eta &= \cos(\theta) \\
y_\xi &= 0 \\
y_\eta &= \sin(\theta)
\end{aligned}
$$

Governing Equations

Untransformed

We define a scalar potential function ψ such that the velocities $\vec{\mathbf{V}} = [u, v]^T$ can be expressed as:

$$
\begin{aligned}
\frac{\partial \psi}{\partial y} &= u \\
\frac{\partial \psi}{\partial x} &= -v
\end{aligned}
$$

We also define the vorticity as the curl of the velocity:

$$\zeta = \nabla \times \vec{\mathrm{V}}$$

For 2D, this reduces to:

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

The evolution equations for both as follows:

Vorticty Transport Equation:

$$
\begin{aligned}
\frac{D\zeta}{Dt} &= \nu \nabla^2 \zeta \\
\frac{\partial \zeta}{\partial t} + u\frac{\partial \zeta}{\partial x} + v\frac{\partial \zeta}{\partial y} &= \nu \left(\frac{\partial^2 \zeta}{\partial x^2} + \frac{\partial^2 \zeta}{\partial^2 y} \right)
\end{aligned}
$$

Poisson's equation for ψ

$$
\begin{aligned}
\nabla^2 \psi &= -\zeta \\
\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} &= -\zeta
\end{aligned}
$$

Transformed equations:

Vorticty Transport Equation:

For the vorticity transport equation, we use **Semi-Lagrangian Advection** to solve the streaming portion. As such the non-viscous contribution is evalutated in the physical domain while the viscous contribution ($\nabla^2\zeta$) is evaluated in the (ξ, η) domain.

$$\frac{D\zeta}{Dt} = \nabla^2\zeta = \left(\xi_x^2 + \xi_y^2\right)\frac{\partial^2\zeta}{\partial\xi^2} + \left(\eta_x^2 + \eta_y^2\right)\frac{\partial^2\zeta}{\partial\eta^2} + 2\left(\xi_x\eta_x + \xi_y\eta_y\right)\frac{\partial^2\zeta}{\partial\xi\partial\eta}$$

Substituting the values of the coordinate derivatives:

$$\nabla^2\zeta = \csc^2(\theta)\,\frac{\partial^2\zeta}{\partial\xi^2} + \csc^2(\theta)\,\frac{\partial^2\zeta}{\partial\eta^2} - 2\cot(\theta)\,\csc(\theta)\,\frac{\partial^2\zeta}{\partial\xi\partial\eta}$$

An operator splitting approach is also explored in the implementation.

$$\begin{aligned} \zeta^{n*} - \zeta^n &= \nu\Delta t\,\nabla^2\zeta^n \\ \zeta^{n+1} &= \text{advect}(\zeta^{n*}, \mathbf{V}, \Delta t) \end{aligned}$$

> ⚠ **Note**
>
> In operator splitting, it turns out that viscocity must be applied ***before*** advecting the vorticity. Else the effects cause incorrect outputs. This was a major source of mystery and frustration until discovered

Poisson's equation for ψ

$$\begin{aligned} \nabla^2\psi &= -\zeta \\ \left(\xi_x^2 + \xi_y^2\right)\frac{\partial^2\psi}{\partial\xi^2} + \left(\eta_x^2 + \eta_y^2\right)\frac{\partial^2\psi}{\partial\eta^2} + 2\left(\xi_x\eta_x + \xi_y\eta_y\right)\frac{\partial^2\psi}{\partial\xi\partial\eta} &= -\zeta \\ \csc^2(\theta)\,\frac{\partial^2\psi}{\partial\xi^2} + \csc^2(\theta)\,\frac{\partial^2\psi}{\partial\eta^2} - 2\cot(\theta)\,\csc(\theta)\,\frac{\partial^2\psi}{\partial\xi\partial\eta} &= -\zeta \end{aligned}$$

Boundary Conditions

Untransformed

ψ

Let $\vec{t}$ and $\vec{n}$ denote the vectors along the wall and perpendicular to the wall respectively. Since the normal velocity (inflow) into the wall is zero, the derivative is the tangential direction will be zero.

$$\vec{\mathbf{V}} \cdot \hat{n} = 0 \implies \frac{\partial \psi}{\partial \vec{t}} = 0$$

This implies that the potential at all connected walls are the same. We can set this constant equal to zero for convenience.

$$\therefore \psi|_{\partial D} = 0$$

where, D is the domain and ∂D is the boundary (the walls)

ζ

We obtain the boundary values of ζ from the neighbourhood values of ψ
Let us consider the top wall of an unskewed domain moving at a velocity of u_0 towards the right.
Let $\psi_{i,N}$ be the point at the boundary at $j = N$ and $\psi_{i,N-1}$ be the point immediately below it.
Then,

$$\psi_{i,N-1} = \psi_{i,N} - \Delta y \left.\frac{\partial \psi}{\partial y}\right|_{i,N} + \frac{\Delta y^2}{2} \left.\frac{\partial^2 \psi}{\partial y^2}\right|_{i,N} + \mathcal{O}(\Delta y^3)$$

We can then substitute for:

$$\left.\frac{\partial \psi}{\partial y}\right|_{i,N} = u_{i,N} = u_0$$

We also know that:

$$\zeta_{i,N} = \cancelto{0}{\left.\frac{\partial v}{\partial x}\right|_{i,N}} - \left.\frac{\partial u}{\partial y}\right|_{i,N} = -\left.\frac{\partial^2 \psi}{\partial y^2}\right|_{i,N}$$

Substituting for the first and second derivatives in the initial fourier series expansion, we obtain:

$$\psi_{i,N-1} = \psi_{i,N} - \Delta y\, u_0 - \frac{\Delta y^2}{2} \zeta_{i,N} + \mathcal{O}(\Delta y^3)$$

From which we get the vorticity as:

$$\zeta_{i,N} = -\frac{2}{\Delta y^2}(\psi_{i,N-1} - \psi_{i,N} + \Delta y\, u_0) + \mathcal{O}(\Delta y)$$

Since the value of ψ at walls is set to zero, we simplify the above as follows:

$$\zeta_{i,N} = -\frac{2}{\Delta y^2}(\psi_{i,N-1} + \Delta y\, u_0) + \mathcal{O}(\Delta y)$$

Similarly we obtain the values of vorticity at other walls:

$$\zeta_{i,1} = -\frac{2}{\Delta y^2}(\psi_{i,2})$$
$$\zeta_{1,j} = -\frac{2}{\Delta x^2}(\psi_{2,j})$$
$$\zeta_{M,j} = -\frac{2}{\Delta x^2}(\psi_{M-1,j})$$

Transformed

ψ

The value of psi does not depend on derivatives as as such the condition remains unchanged.

ζ

Approach 1 - vertical walls

Let us take the case for the top wall, the area being a (1×1) square in the computational domain. Then,

$$\psi_{i,N-1} = \psi_{i,N} - \Delta\eta \left.\frac{\partial \psi}{\partial \eta}\right|_{i,N} + \frac{\Delta\eta^2}{2} \left.\frac{\partial^2 \psi}{\partial \eta^2}\right|_{i,N} + \mathcal{O}(\Delta\eta^3)$$

Analysing the first derivative term,

$$\left.\frac{\partial \psi}{\partial \eta}\right|_{i,N} = x_\eta \cancelto{0}{\left.\frac{\partial \psi}{\partial x}\right|_{i,N}} + y_\eta \left.\frac{\partial \psi}{\partial y}\right|_{i,N} = \sin(\theta) \left.\frac{\partial \psi}{\partial y}\right|_{i,N} = \sin(\theta) u_0$$

Looking at the second derivative term,

$$\frac{\partial^2 \psi}{\partial \eta^2} = x_\eta^2 \frac{\partial^2 \psi}{\partial x^2} + y_\eta^2 \frac{\partial^2 \psi}{\partial y^2} + 2\, x_\eta y_\eta \frac{\partial^2 \psi}{\partial x \partial y}$$

Note, is is implicitly assumed that the higher order derivatives of the coordinates are zero.
Additionally,

$$\left.\frac{\partial^2 \psi}{\partial x \partial y}\right|_{i,N} = \left.\frac{\partial}{\partial x}\left(\frac{\partial \psi}{\partial y}\right)\right|_{i,N} = \left.\frac{\partial u}{\partial x}\right|_{i,N} = 0$$

And:

$$\left.\frac{\partial^2\psi}{\partial x^2}\right|_{i,N} = 0 \quad \because \quad \psi = \text{constant}$$

This leaves only the second derivative in y. Substituting for vorticity from earlier equation as:

$$\zeta_{i,N} = -\left.\frac{\partial^2\psi}{\partial y^2}\right|_{i,N}$$

We obtain the relation between the second derivative and vorticity

$$\left.\frac{\partial^2\psi}{\partial \eta^2}\right|_{i,N} = -\sin^2(\theta)\zeta_{i,N}$$

This allows us to finally obtain the relation:

$$\zeta_{i,N} = -\frac{2}{\Delta\eta^2\sin^2(\theta)}(\psi_{i,N-1} + \Delta\eta\,\sin(\theta)\,u_0) + \mathcal{O}(\Delta\eta)$$

For the bottom wall, we can obtain a similar relation:

$$\zeta_{i,1} = -\frac{2}{\Delta\eta^2\sin^2(\theta)}\,\psi_{i,2} + \mathcal{O}(\Delta\eta)$$

Approach 2 - Non-horizontal walls

In this process we attach a local coordinate system separate from the (ξ, η) and (x, y) coordinate system.
Let us define two vectors:

$$\vec{t} = [\cos(\theta), \sin(\theta)]^T$$

It is the vector aligning tangent to the left and right boundary. We also define an $\vec{n}$ direction perpendicular to $\vec{t}$

$$\vec{n} = [\sin(\theta), -\cos(\theta)]^T$$

From this we can obtain $x\hat{x} + y\hat{y}$ from $(t\hat{t} + n\hat{n})$:

$$x = t\cos(\theta) + n\sin(\theta)$$
$$y = t\sin(\theta) - n\cos(\theta)$$

$$t = x\cos(\theta) + y\sin(\theta)$$
$$n = x\sin(\theta) - y\cos(\theta)$$

$$t = \xi\cos(\theta) + \eta\left(1 + \cos^2(\theta)\right)$$
$$n = -\xi\sin(\theta)$$

And we obtain the derivatives of t and n

$$t_x = \cos(\theta), \quad t_y = \sin(\theta), \quad t_\xi = \cos(\theta), \quad t_\eta = 1 + \cos^2(\theta)$$
$$n_x = \sin(\theta), \quad n_y = -\cos(\theta), \quad n_\xi = -\sin(\theta), \quad n_\eta = 0$$

Since $\hat{t}\cdot\hat{t} = \hat{n}\cdot\hat{n} = 1$

$$-\zeta = \nabla^2\psi = \frac{\partial^2\psi}{\partial t^2} + \frac{\partial^2\psi}{\partial n^2}$$

At the boundary(left wall), since ψ is constant($= 0$) along the t coordinate at the walls:

$$-\zeta_{1,j} = \cancelto{0}{\left.\frac{\partial^2\psi}{\partial t^2}\right|_{1,j}} + \left.\frac{\partial^2\psi}{\partial n^2}\right|_{1,j}$$

In addition the velocities at the walls is zero,

$$\left.\frac{\partial\psi}{\partial t}\right|_{1,j} = 0, \quad \left.\frac{\partial\psi}{\partial n}\right|_{1,j} = 0$$

Evaluating ψ at the left wall:

$$\psi_{2,j} = \psi_{1,j} + \Delta\xi\left.\frac{\partial\psi}{\partial\xi}\right|_{1,j} + \frac{\Delta\xi^2}{2}\left.\frac{\partial^2\psi}{\partial\xi^2}\right|_{1,j} + \mathcal{O}(\Delta\xi^3)$$

Swicthing variables,

$$\left.\frac{\partial\psi}{\partial\xi}\right|_{1,j} = \cancelto{0}{t_\xi\left.\frac{\partial\psi}{\partial t}\right|_{1,j}} + n_\xi\left.\frac{\partial\psi}{\partial n}\right|_{1,j}$$

$$\left.\frac{\partial^2\psi}{\partial\xi^2}\right|_{1,j} = \cancelto{0}{t_\xi^2\left.\frac{\partial^2\psi}{\partial t^2}\right|_{1,j}} + n_\xi^2\left.\frac{\partial^2\psi}{\partial n^2}\right|_{1,j} + \cancelto{0}{2\,t_\xi n_\xi\left.\frac{\partial^2\psi}{\partial t\partial n}\right|_{1,j}}$$

$$\left.\frac{\partial\psi}{\partial\xi}\right|_{1,j} = n_\xi\left.\frac{\partial\psi}{\partial n}\right|_{1,j}$$

$$\left.\frac{\partial^2\psi}{\partial\xi^2}\right|_{1,j} = n_\xi^2\left.\frac{\partial^2\psi}{\partial n^2}\right|_{1,j}$$

Substituting all the above into our series expansion of ψ at the left wall,

$$\psi_{2,j} = \frac{\Delta\xi^2}{2}\, n_\xi^2\, \frac{\partial^2\psi}{\partial n^2}\bigg|_{1,j} + \mathcal{O}(\Delta\xi^3)$$

Isolating ζ:

$$\zeta_{1,j} = -\frac{2}{\Delta\xi^2\, n_\xi^2}\, \psi_{2,j} + \mathcal{O}(\Delta\xi)$$

We similarly obtain the value for the right wall as:

$$\zeta_{N,j} = -\frac{2}{\Delta\xi^2\, n_\xi^2}\, \psi_{N-1,j} + \mathcal{O}(\Delta\xi)$$

FDM discretization

Vorticity Transport

We apply viscocity first and then advect it.

$$\begin{aligned} \zeta^{n*} - \zeta^n &= \nu\Delta t\, \nabla^2\zeta^n \\ \zeta^{n+1} &= \mathrm{advect}(\zeta^{n*}, \mathbf{V}, \Delta t) \end{aligned}$$

Applying Viscosity:

We apply a five point central difference to approximate the viscocity.

$$\begin{aligned} \nu\nabla^2\zeta_{i,j} = \nu\Bigg[& (\xi_x^2 + \xi_y^2)\, \frac{\zeta_{i+1,j} - 2\zeta_{i,j} + \zeta_{i-1,j}}{\Delta\xi^2} \\ & + (\eta_x^2 + \eta_y^2)\, \frac{\zeta_{i,j+1} - 2\zeta_{i,j} + \zeta_{i,j-1}}{\Delta\eta^2} \\ & + 2\,(\xi_x\eta_x + \xi_y\eta_y)\, \frac{\zeta_{i+1,j+1} - \zeta_{i-1,j+1} - \zeta_{i+1,j-1} + \zeta_{i-1,j-1}}{4\,\Delta\xi\,\Delta\eta} \Bigg] \end{aligned}$$

$$\nu\nabla^2\zeta_{i,j} = \nu\left(\frac{\zeta_{i+1,j} - 2\zeta_{i,j} + \zeta_{i-1,j}}{\Delta x^2} + \frac{\zeta_{i,j+1} - 2\zeta_{i,j} + \zeta_{i,j-1}}{\Delta y^2} \right)$$

Interpolation

Since we have the viscosity at only the nodal values, we use bi-linear interpolation to get values between nodal points

$$\alpha = \frac{\xi - \xi_i}{\Delta\xi}, \qquad \beta = \frac{\eta - \eta_j}{\Delta\eta}$$

$$\zeta(\mathbf{x}) = (1-\alpha)(1-\beta)\,\zeta_{i,j} + (1-\alpha)\beta\,\zeta_{i,j+1} + \alpha\beta\,\zeta_{i+1,j+1} + \alpha(1-\beta)\,\zeta_{i+1,j}$$

Since we require velocity at non-nodal values, we also interpolate velocity.

$$\mathbf{u}(\mathbf{x}) = (1-\alpha)(1-\beta)\,\mathbf{u}_{i,j} + (1-\alpha)\beta\,\mathbf{u}_{i,j+1} + \alpha\beta\,\mathbf{u}_{i+1,j+1} + \alpha(1-\beta)\,\mathbf{u}_{i+1,j}$$

Advection

We use semi-Lagrangian advection to transport viscosity

$$\begin{aligned}
\mathbf{k}_1 &= -\mathbf{u}(\mathbf{x}^n) \\
\mathbf{k}_2 &= -\mathbf{u}\left(\mathbf{x}^n + \frac{\Delta t}{2}\mathbf{k}_1\right) \\
\mathbf{k}_3 &= -\mathbf{u}\left(\mathbf{x}^n + \frac{\Delta t}{2}\mathbf{k}_2\right) \\
\mathbf{k}_4 &= -\mathbf{u}(\mathbf{x}^n + \Delta t\,\mathbf{k}_3)
\end{aligned}$$

$$\mathbf{x}^{\,\text{back}} = \mathbf{x}^n + \frac{\Delta t}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4)$$

$$\zeta^{n+1}(\mathbf{x}^n) = \zeta^{n*}(\mathbf{x}^{\,\text{back}})$$

Solving Poisson's equation.

Once the vorticity is updated, we need to solve for the stream function using the following relation:

$$\nabla^2\psi = -\zeta$$

$$\begin{aligned}
\psi_{i,j}^{(k+1)} = \frac{1}{2(a_1+a_3)}\Bigg[& a_1\left(\psi_{i-1,j}^{(k)} + \psi_{i+1,j}^{(k)}\right) \\
& + a_2\left(\psi_{i+1,j+1}^{(k)} - \psi_{i-1,j+1}^{(k)} - \psi_{i+1,j-1}^{(k)} + \psi_{i-1,j-1}^{(k)}\right) \\
& + a_3\left(\psi_{i,j-1}^{(k)} + \psi_{i,j+1}^{(k)}\right) + \omega_{i,j}\,\Delta\xi^2\Bigg]
\end{aligned}$$

where,

$$a_1 = \xi_x^2 + \xi_y^2, \quad a_2 = \tfrac{1}{2}(\xi_x\eta_x + \xi_y\eta_y)\frac{\Delta\xi}{\Delta\eta}, \quad a_3 = (\eta_x^2 + \eta_y^2)\left(\frac{\Delta\xi}{\Delta\eta}\right)^2$$

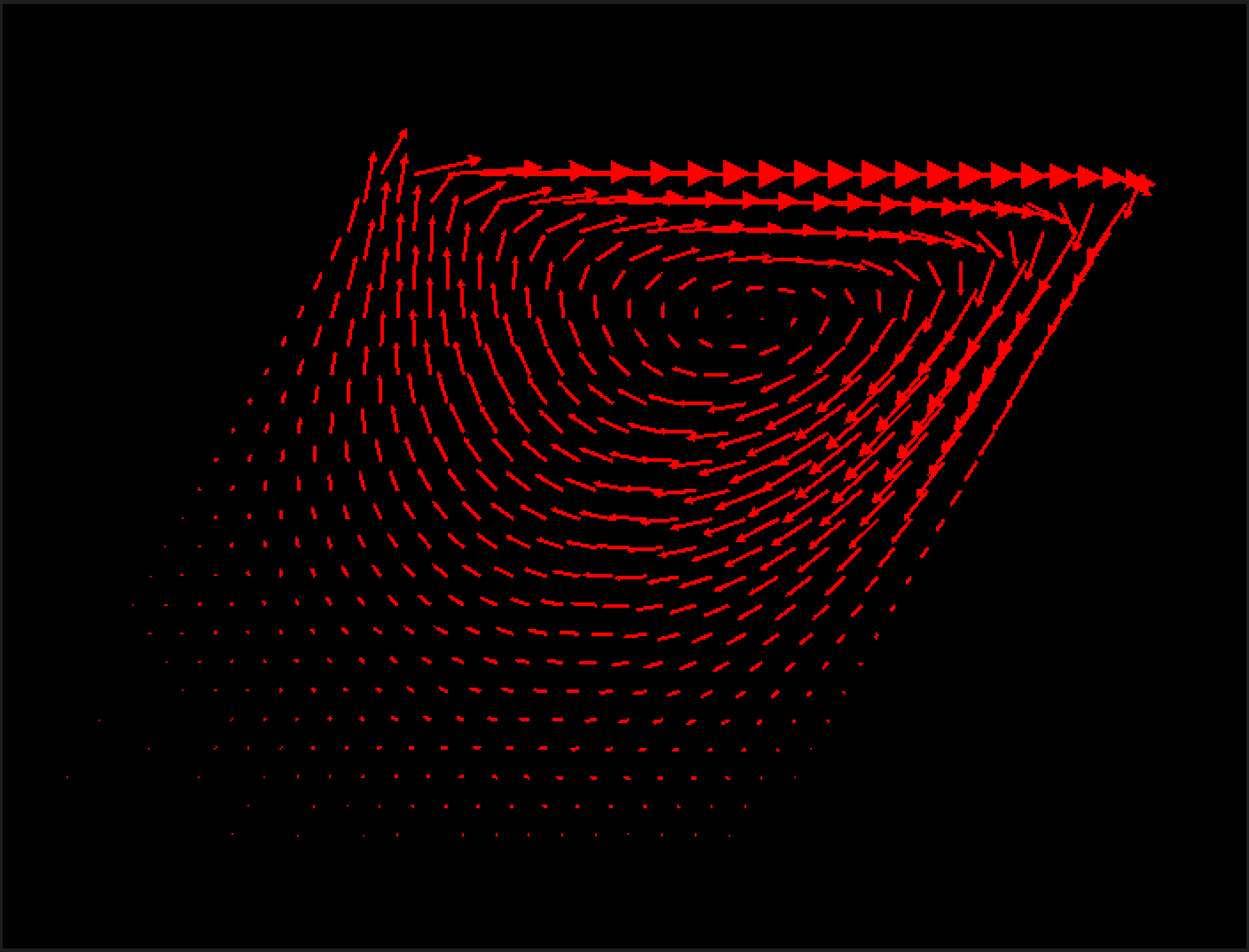
Results

Line contour plots are plotted along with centerline velocity plots for Reynold's numbers of 100 and 1000.

Note that the paper referenced does not have the centerline plot for Re=100 for skewed cavity but instead has it for Re=1000.

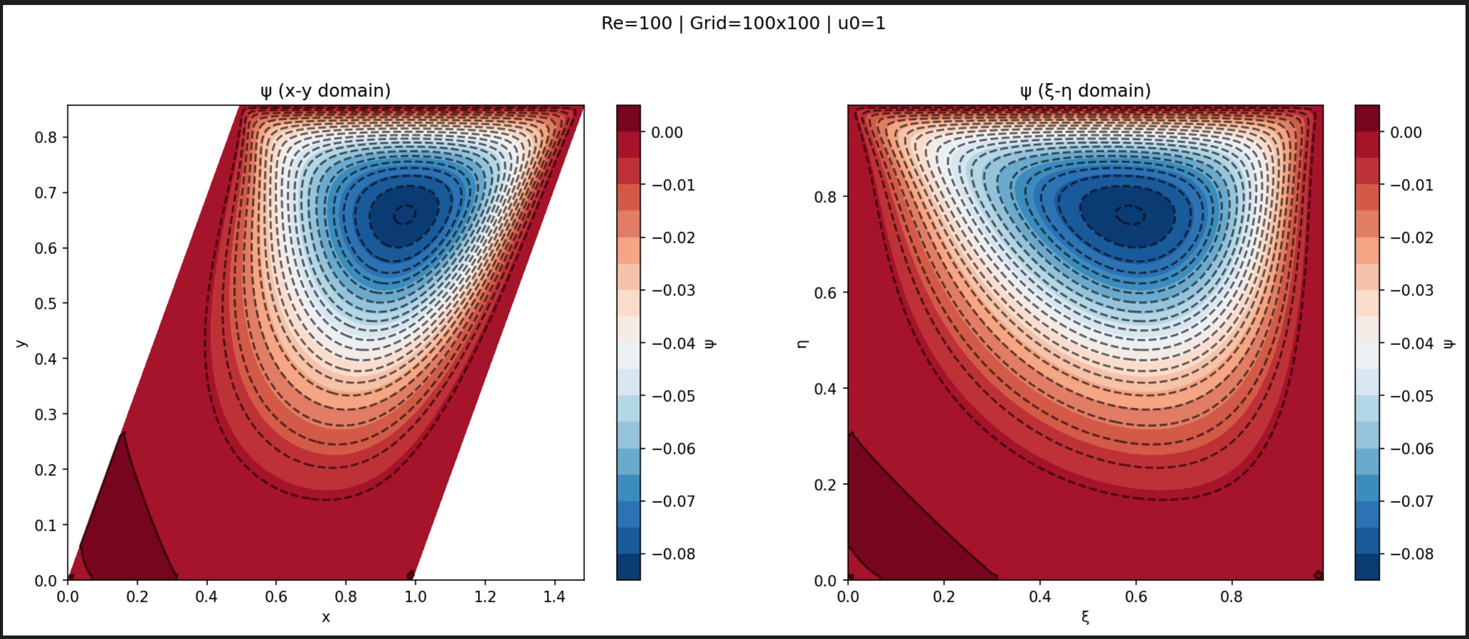
The domain is skewed at an angle of 45°

Re = 100

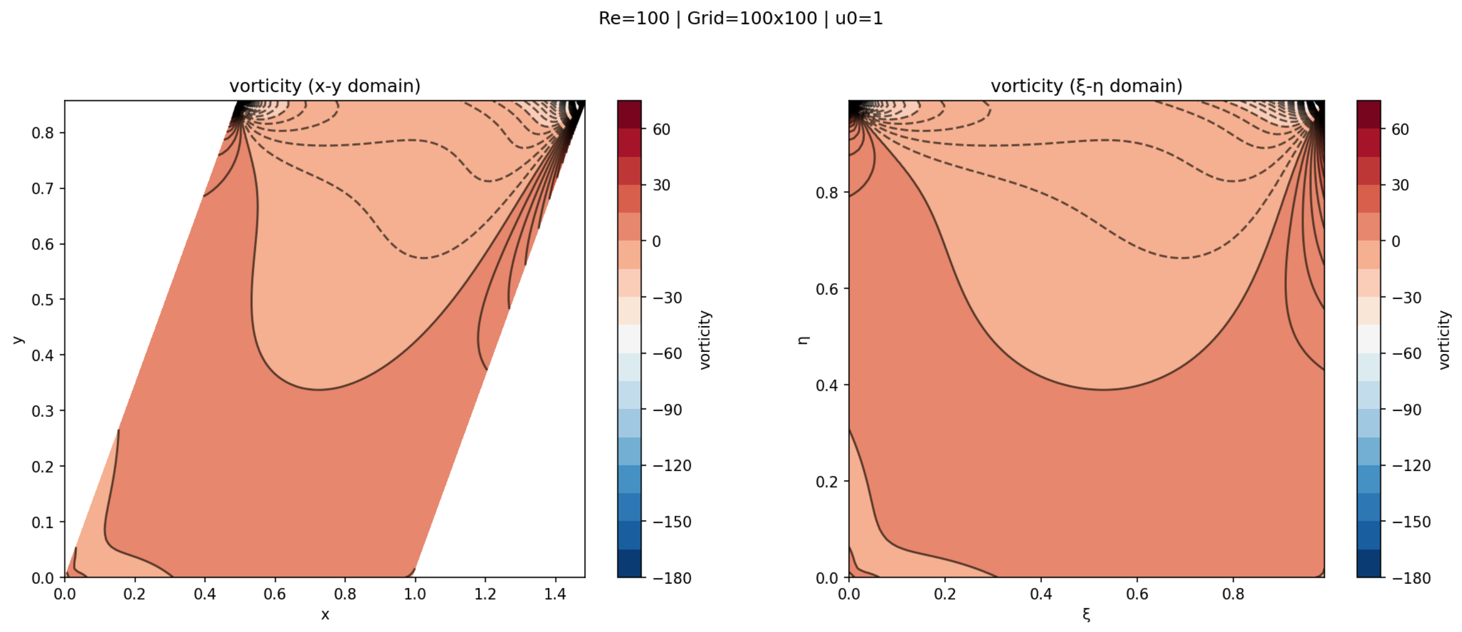


Contour Plots

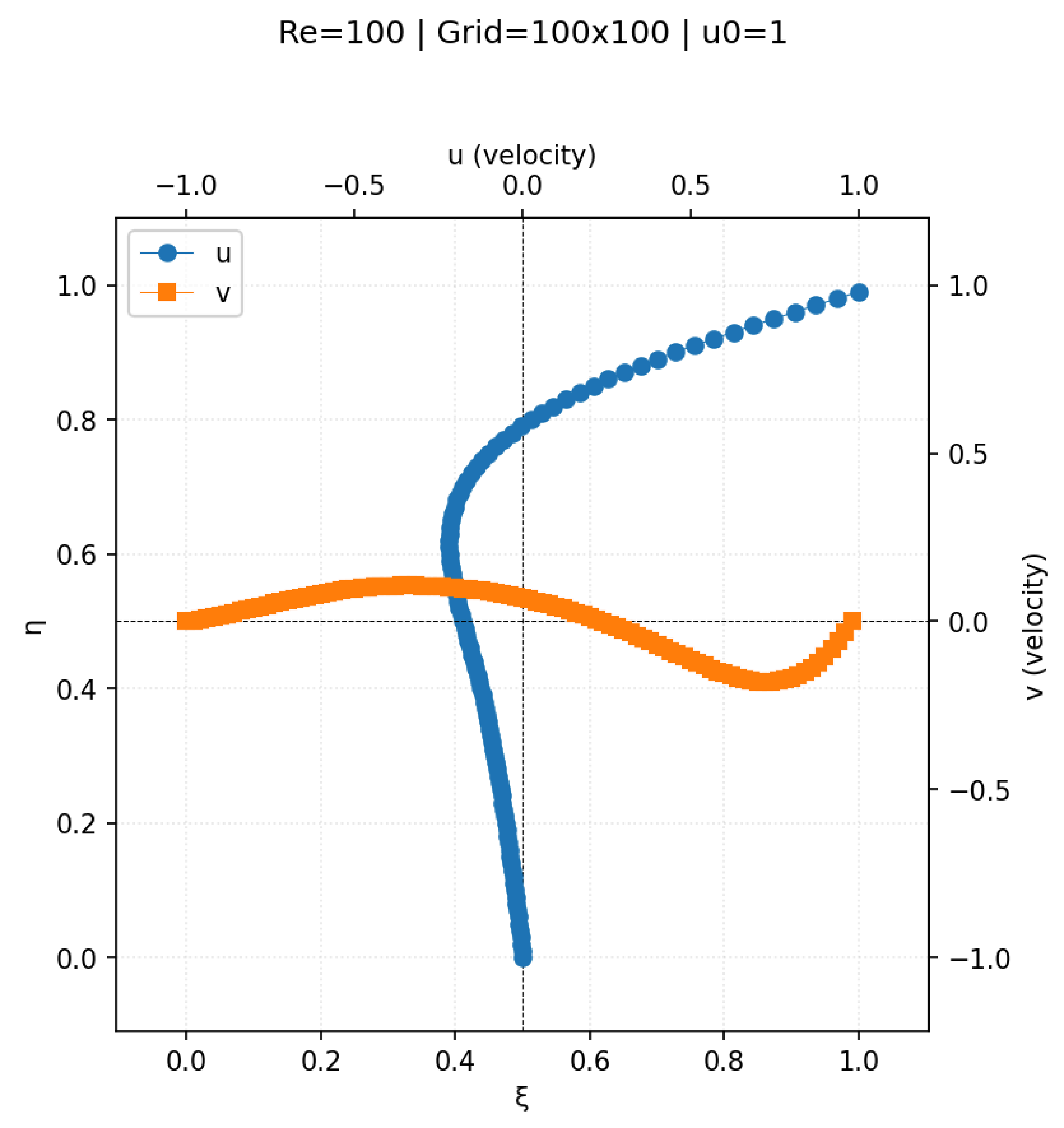
Stream function ψ



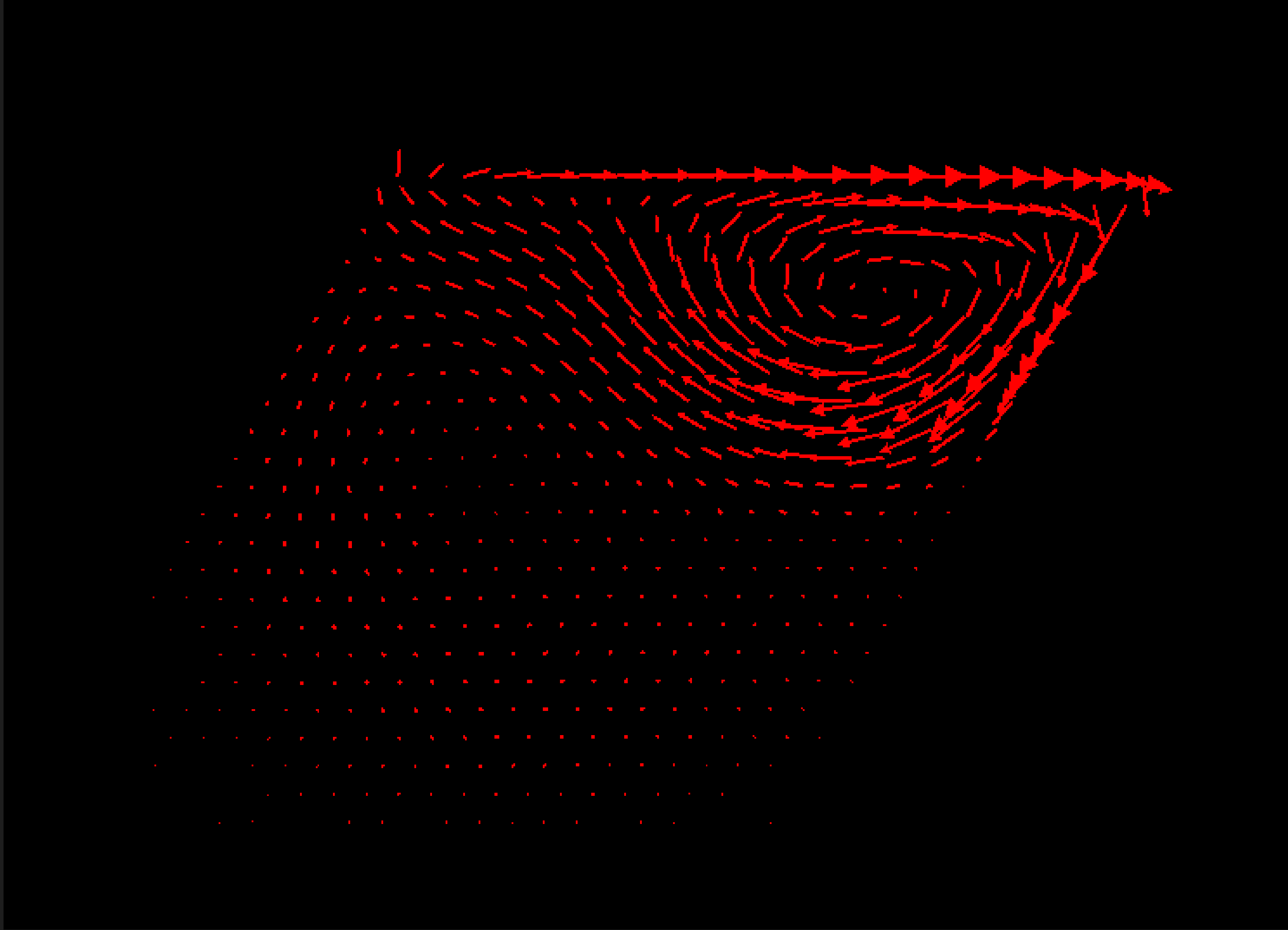
Vorticity ζ



Centerline velocities

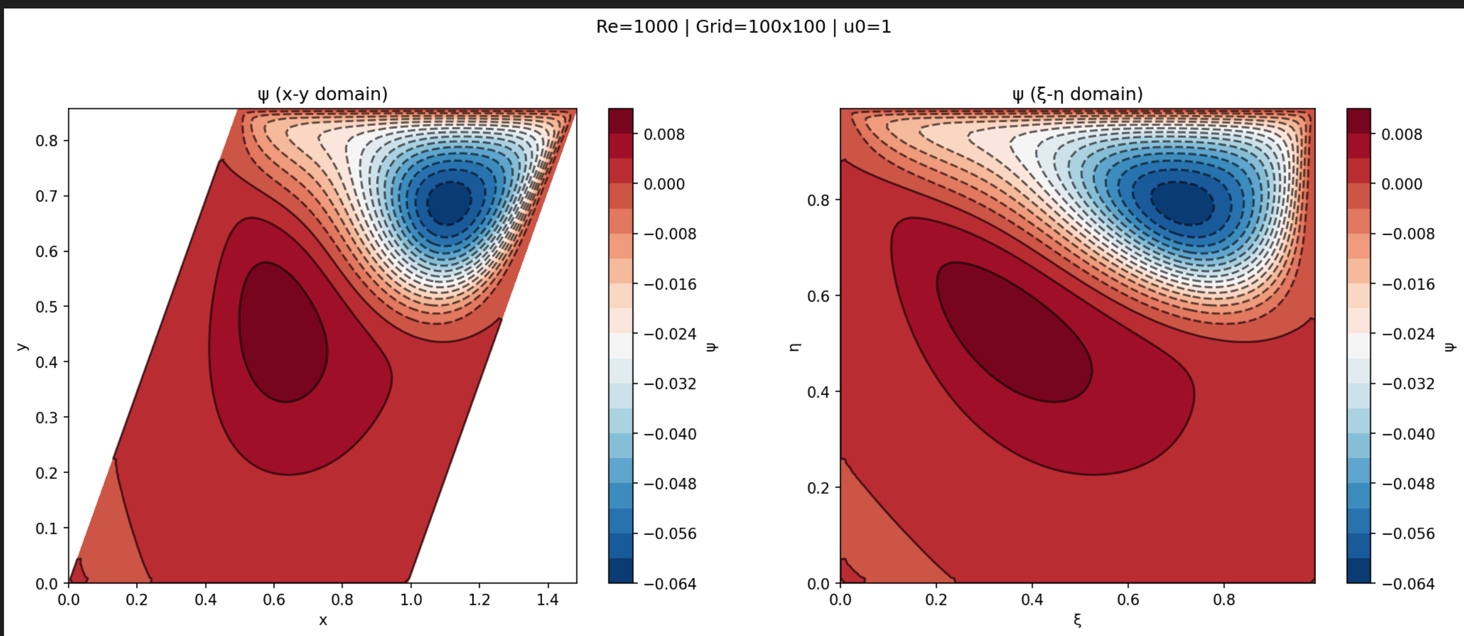

Re=100 | Grid=100x100 | u0=1
u (velocity)
−1.0
−0.5
0.0
0.5
1.0
u
v
1.0
0.8
0.6
0.4
0.2
0.0
η
1.0
0.5
0.0
−0.5
−1.0
v (velocity)
0.0
0.2
0.4
0.6
0.8
1.0
ξ

Re = 1000

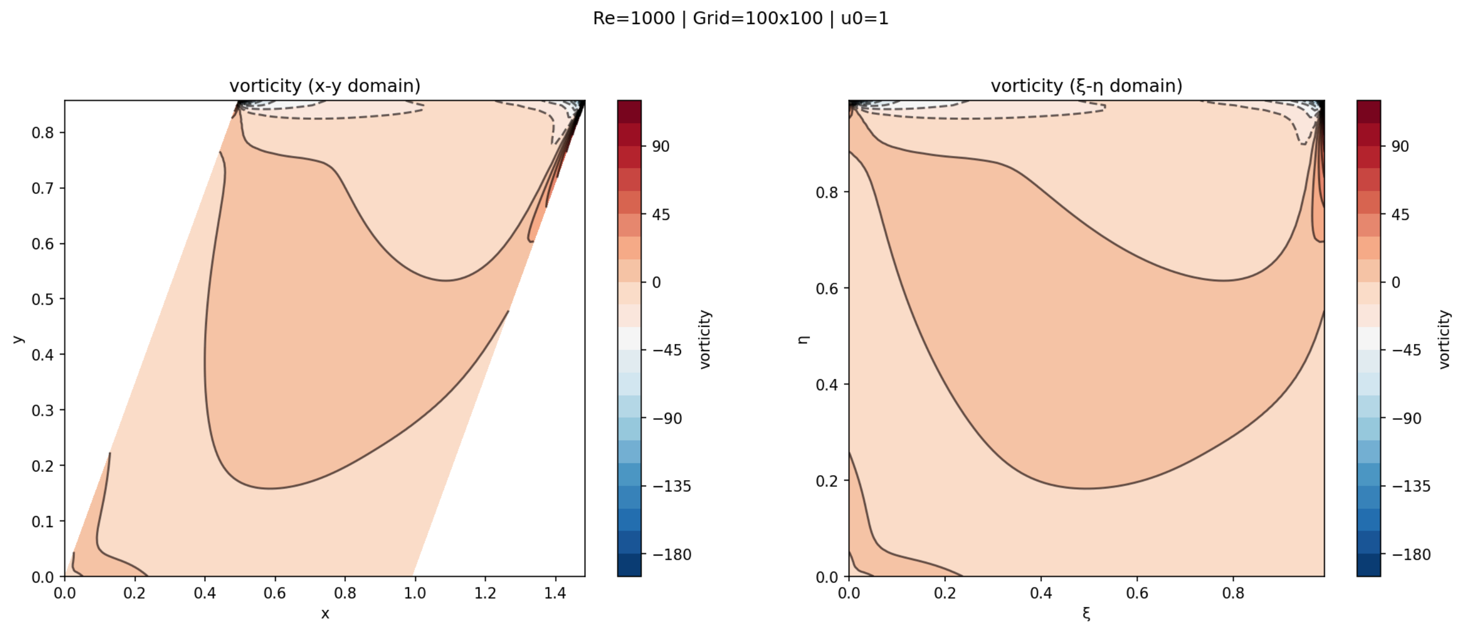


Contour plots

Steam function ψ

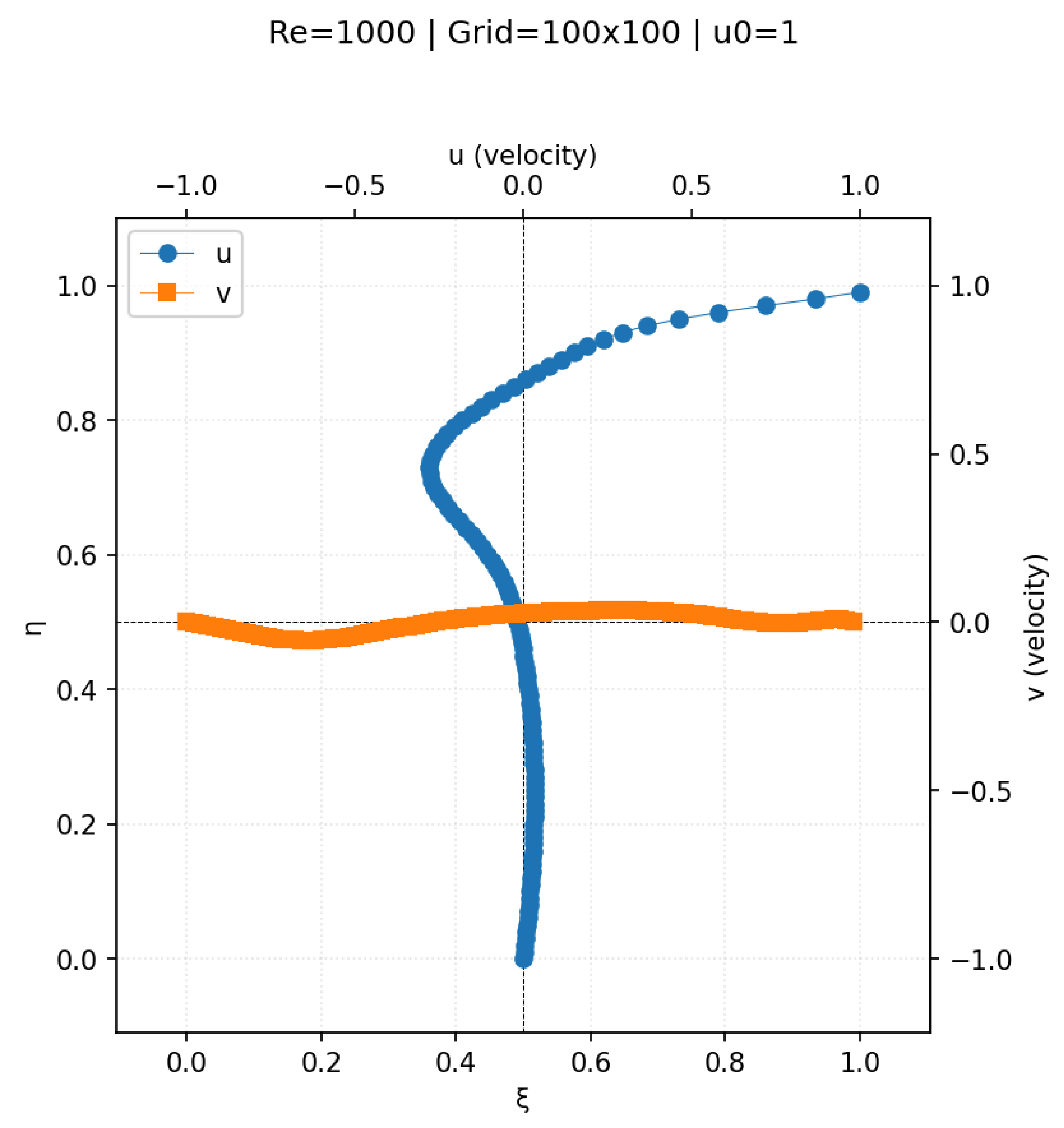


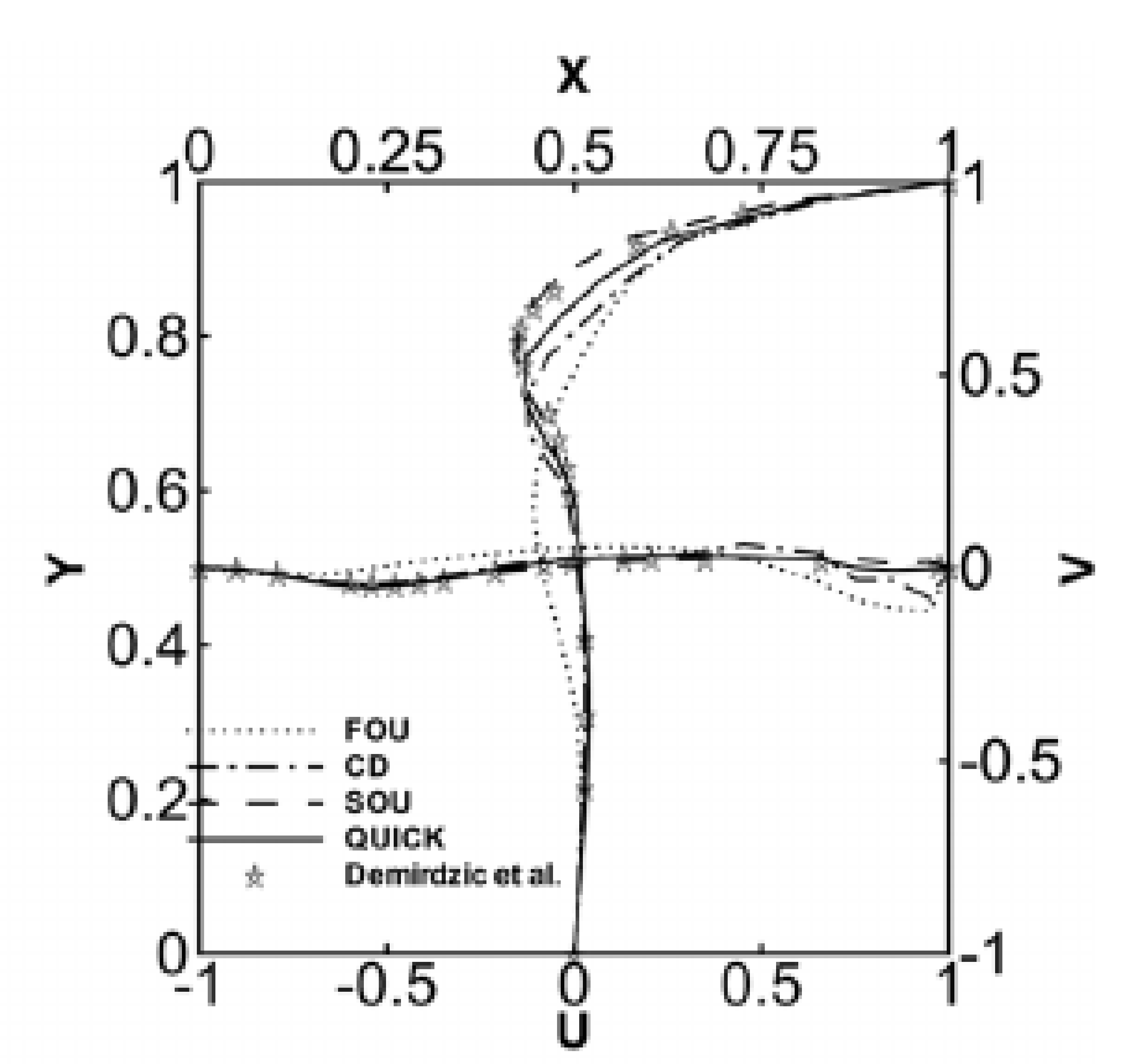
Vorticity ζ



Centerline velocities

The below graph using the same scaling as the reference paper whose image is attach below this one


Re=1000 | Grid=100x100 | u0=1
u (velocity)
−1.0
−0.5
0.0
0.5
1.0
u
v
η
ξ
v (velocity)
0.0
0.2
0.4
0.6
0.8
1.0



(b) Skew cavity (β=45^0)

Conclusion

Here a vorticity-stream function based solver is presented for a skewed(45 degrees) cavity. The results agree closely with the reference paper.
All code is open source and available on github at https://github.com/Sam-MARTis/Voriticy-Stream-Function-Fluid-Sim

References

1. Sachin B. Paramane & Atul Sharma (2008) Consistent Implementation and Comparison of FOU, CD, SOU, and QUICK Convection Schemes on Square, Skew, Trapezoidal, and Triangular Lid-Driven Cavity Flow, Numerical Heat

Transfer, Part B: Fundamentals: An International Journal of Computation and Methodology, 54:1, 84-102, DOI: 10.1080/10407790802122519